

# Hedgehog Signaling in Germ Cell Migration

Girish Deshpande,<sup>1</sup> Lisa Swanhart, Phyllis Chiang, and Paul Schedl

Department of Molecular Biology  
Princeton University  
Princeton, New Jersey 08544

## Summary

The primitive gonad of the *Drosophila* embryo is formed from two cell types, the somatic gonad precursor cells (SGPs) and the germ cells, which originate at distant sites. To reach the SGPs the germ cells must undergo a complex series of cell movements. While there is evidence that attractive and repulsive signals guide germ cell migration through the embryo, the molecular identity of these instructive molecules has remained elusive. Here, we present evidence suggesting that *hedgehog* (*hh*) may serve as such an attractive guidance cue. Misexpression of *hh* in the soma induces germ cells to migrate to inappropriate locations. Conversely, cell-autonomous components of the *hh* pathway appear to be required in the germline for proper germ cell migration.

## Introduction

The primitive gonad in *Drosophila* embryos is composed of two distinct cell types—the germ cells, which are derived from a special group of cells called the pole cells, and the somatic gonadal precursor cells, which are derived from the mesoderm (Boyle and DiNardo, 1995). Pole cells have a number of unique properties which distinguish them from somatic cells (Williamson and Lehmann, 1996). They are specified by maternal determinants localized in pole plasm at the posterior end of the embryo (St. Johnston, 1993; Williamson and Lehmann, 1996). These maternal determinants are assembled at the pole during oogenesis under the direction of the posterior organizing factor Oskar (Ephrussi and Lehmann, 1992; Smith et al., 1992). The pole cells cellularize precociously around nuclear cycle 10, when a few nuclei migrate from the center of the embryo into the pole plasm (Zalokar and Erk, 1976). Though the pole cells continue to divide after they are formed, their rate of mitosis is much slower than the surrounding somatic nuclei, and they stop dividing altogether by the cellular blastoderm stage. Pole cells also differ from the soma in that they turn off transcription and remain transcriptionally quiescent until stage 9–10 of embryogenesis (Kobayashi et al., 1996; Seydoux and Strome, 1999).

While zygotic transcription is not required for pole cell formation, the specification of the somatic gonadal precursor cells (SGP) depends upon a complex interplay of zygotically active patterning genes. Combinatorial regulatory interactions between segmentation genes coupled with inductive signals from the dorsal ectoderm

and the CNS midline partition presumptive mesodermal cells into alternative fates—heart, muscle, fat body, or SGPs (Staehling-Hampton et al., 1994; Frasch, 1995; Azpiazu et al., 1996; Azpiazu and Frasch, 1996; Luer et al. 1997). SGPs are formed in three bilateral cell clusters within the *even-skipped* (*eve*) domain in parasegments (PS)10–12 at a position located immediately ventral to the precursors of the visceral mesoderm (Brookman et al., 1992; Broihier et al., 1998). Mesodermal cells situated at the identical positions in the more anterior parasegments, PS4–9, form the primary fat body cell clusters. Because of the similarities in relative A-P and D-V positions in each parasegment and in their specification process, the fat body primordia in PS4–9 and SGP primordia in PS10–12 are regarded as homologous structures (Riechmann et al., 1998, Moore et al., 1998a).

In addition to *eve*, at least six genes are known to play important roles in the choice of SGP cell fate. These are *decapentaplegic* (*dpp*), *tinman* (*tin*), *engrailed* (*en*), *hedgehog* (*hh*), *wingless* (*wg*), and *Abdominal-A* (*abd-A*). *dpp* functions to induce and maintain *tin* expression in the presumptive SGP cells. It is thought that early *tin* expression establishes competence for SGP formation, while *Abd-A*, through its repression of the *serpent* (*srp*) gene, prevents the presumptive SGP cells from assuming fat body identity (Sam et al., 1996). *en* and *hh*, which are expressed in the SGP cells, then function to maintain competence for SGP cell formation. *wg*, which is not expressed in the SGP cells, is also required for SGP formation because it functions to maintain *en* expression (Riechmann et al., 1998; Moore et al., 1998b).

The initial specification of the SGP cells is marked by the expression of the *cliff* gene in two bilateral clusters of 9–12 cells in PS10 through 12 (Boyle et al., 1997). Soon thereafter, migrating pole cells associate with these SGP cell clusters. The germ cells remain associated with the SGPs during germ band retraction and together, the two groups of cells move anteriorly and then coalesce into the embryonic gonad. To reach the SGP clusters, the pole cells undergo a complex pattern of migration. When the pole cells are first formed at the syncytial blastoderm stage, they reside outside of the embryo at the posterior pole. The pole cells remain segregated from the soma after cellularization and through much of gastrulation. During the initial phase of gastrulation, they are carried anteriorly along the dorsal surface of the embryo and then internalized into the center of the embryo with the posterior midgut invagination. Subsequently, the germ cells traverse the posterior midgut endoderm, migrating dorsally to the overlying mesoderm. After reaching the mesoderm, the germ cells split into two clusters, which migrate laterally until they establish contacts with the somatic gonadal precursor cells (SGPs) (Warrior, 1994; Jaglarz and Howard, 1995; Moore et al., 1998b).

In order to generate a functional gonad, the migrating germ cells need to be guided away from the posterior midgut toward the SGPs. After migration, the germ cells must be able to recognize the SGPs, and form a stable association. The process of migration is thought to involve repulsive and attractive cues that guide the migrat-

<sup>1</sup> Correspondence: gdeshpande@molbio.princeton.edu

ing germ cells away from the posterior midgut toward the SGPs (Howard, 1998). However, little is known about the nature of the cues or how they function to direct the migration of the germ cells. Similarly, the mechanisms involved in initiating and sustaining interactions between the germ cells and the SGPs are not understood.

We wondered if the genes involved in the specification of the somatic gonadal precursor cells might be involved in generating guidance cues for germ cells or facilitating contacts between germ cells and SGPs. Among the several genes that are critical for the specification and differentiation of SGPs, we have focused our attention on hedgehog (Hh) because it is expressed in SGP precursor cells (see below) and can function as a short and long range signaling molecule (Alcedo and Noll, 1997; Ingham, 1998; Johnson and Scott, 1998; Chuang and Kornberg, 2000; Kalderon, 2000). We show here that an Hh-dependent signal exerts an attractive influence on germ cells, directing their migration toward the source of Hh protein. Supporting the idea that Hh produced by the differentiating SGP cells is the attractive signal, we present evidence indicating that components of the *hh* pathway which function in the reception of the *hh* signal are needed in the germline for proper migration.

## Results

### *hh-lacZ* Is Transcribed in SGP Cells

Previous studies have implicated *hh* in the specification of the SGP cells (Riechmann et al., 1998); however, while it is known that *hh* is expressed in the overlying ectoderm, it has not been established whether *hh* is also expressed in SGP cells themselves or in their neighboring mesodermal cells. To address this question, we immunostained embryos carrying a transgene in which *hh* promoter drives *LacZ* expression with antibodies against  $\beta$ -galactosidase (imaged in green) and the SGP marker Clift (imaged in red). As shown in the confocal images in Figure 1,  $\beta$ -galactosidase can be detected in the Clift-positive mesodermal cells in parasegments 10, 11, and 12 in germ band extended embryos. Assuming that the expression pattern of the *hh-lacZ* transgene faithfully mimics the endogenous *hh* gene, this finding indicates that *hh* is expressed in the SGPs. Note that  $\beta$ -galactosidase can also be detected at this stage in Clift-negative mesodermal cells in parasegments anterior to PS10. Based on their homologous position in the mesoderm, these cells are likely to be fat body precursors. This is consistent with the finding that *hh* is required not only for the formation of the somatic gonad but also for the specification of the fat body (Riechmann et al., 1998).

### Ectopically Expressed Hh Can Induce Germ Cell Migration

Since *hh* seems to be expressed in SGPs just at the time when the germ cells begin to exit the midgut, we wondered whether this signaling molecule has an additional function in orchestrating the migration of germ cells through the embryo and/or their association with the SGP cells. If *hh* does play a role in germ cell movement, then it should be possible to alter the migration pattern of these cells by expressing the Hh protein at ectopic sites in the embryo. For this purpose, we took

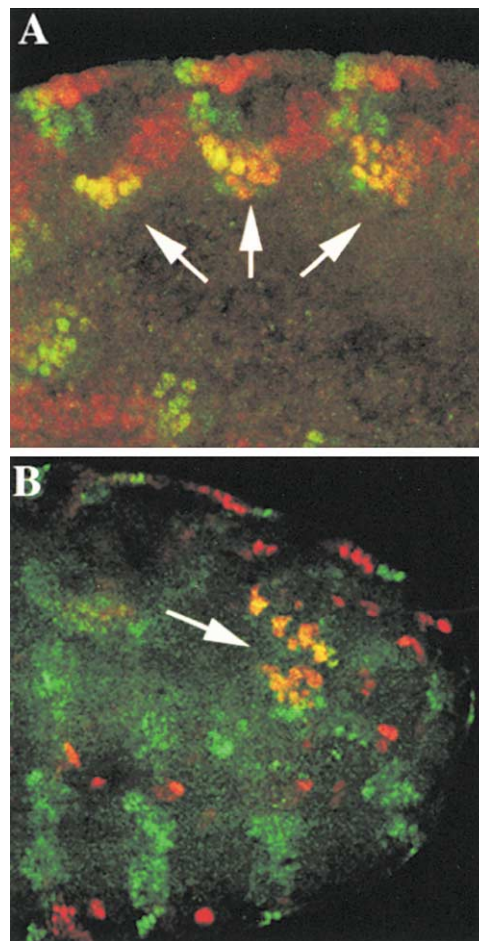


Figure 1. *hh* Is Expressed in SGP Cells

Confocal images of two embryos at stage 10 (A) and 12 (B) stained with  $\beta$ -galactosidase (imaged in green) and Clift (imaged in red) antibodies. Embryos carrying a transgene in which the *hh* promoter drives *LacZ* expression were immunostained with antibodies against  $\beta$ -galactosidase. In order to identify the SGP cells, the embryos were also immunostained with antibodies against Clift. The SGP cells in PS10–12 of the embryo shown here are stained with both  $\beta$ -galactosidase and Clift antibodies (see arrows).

advantage of the UAS-*Gal4* misexpression system (Brand and Perrimon, 1993).

In the first experiment, we induced the expression of Hh protein from the UAS-*hh* transgene in alternate segments using a *hairy-Gal4* driver and then examined the effects on germ cell migration. The UAS-*hh* transgene by itself has no effect on the migration behavior of germ cells, and the typical elongated cluster of Vasa positive germ cells is observed in the mesoderm near the posterior end of stage 13 embryos (Figure 2: Late UAShh/UAShh). While a germ cell cluster is observed at this same position in stage 13 *hairy-Gal4/UAS-hh* embryos, germ cells are also seen at a variety of abnormal locations in the posterior ectoderm (Figure 2: Late). Migration defects are also evident at earlier stages just after the germ cells leave the posterior midgut. Instead of migrating dorsally toward the overlying mesoderm (Figure 2: Early UAShh//+) some of the germ cells in stage 10–11 *hairy-Gal4/UAS-hh* embryos remain associated with the posterior midgut (middle panel) or migrate

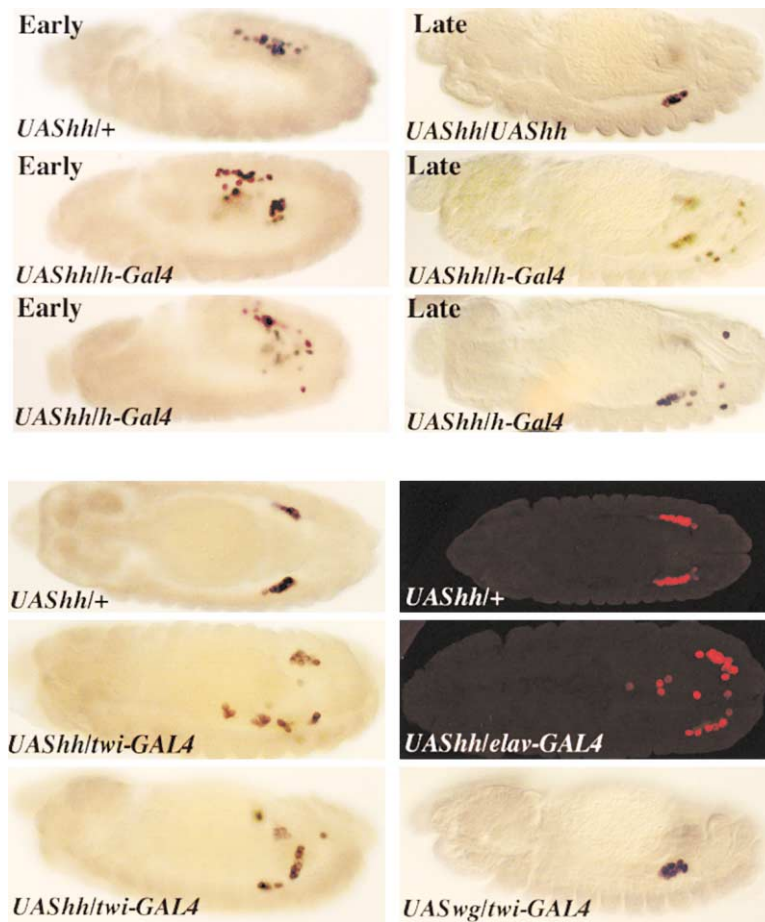


Figure 2. Ectopic Hh Disrupts Germ Cell Migration.

Whole-mount antibody staining of embryos of the indicated genotype with antibodies against the germ-cell-specific Vasa protein. Embryos are oriented anterior to the left, dorsal up. The staining was detected using standard immunohistochemical reaction, except in the case of *elav-GAL4*, where immunofluorescence-based detection was utilized (for details of this experiment, see Experimental Procedures). The three panels on the upper left labeled Early show Vasa-stained stage 11 embryos. The three panels on the upper right labeled Late show Vasa-stained stage 13 embryos. The embryos shown in the bottom left and right panels are all stage 13. The embryos in this figure are representative examples from samples of >50.

in the opposite direction toward the ventral side of the embryo (bottom panel).

Ectopic expression of Hh protein in the CNS using an *elav-Gal4* driver also induces aberrant migration patterns. In stage 13 wild-type or control *UAS-hh* transgene embryos (see Figure 2), the germ cells are typically arranged in two elongated but tight clusters on either side of the ventral midline. While two clusters can usually be discerned in *elav-GAL4/UAS-hh* stage 13 embryos, most of the cells in the clusters are not tightly associated with each other and some are dispersed at distant sites in the posterior mesoderm (Figure 2). In addition, a subset of the germ cells are typically found near the ventral midline in the region just underlying the CNS where expression of Hh from the *elav-GAL4* driver is expected to be highest at this stage of development (see the germ cells in between the two clusters in the *elav-GAL4/UAS-hh* embryo in Figure 2). Perhaps because of differences in timing and/or tissue specificity, Hh expression directed by the *elav-GAL4* driver does not noticeably perturb the early steps in migration when the germ cells have just exited the gut (not shown).

Even more dramatic alterations in germ cell migration are observed when *hh* expression is driven in the mesoderm by a *twist-GAL4* driver. As shown in the stage 12–13 embryos in Figure 2, germ cells are found in several small clusters of 3–6 cells scattered in different segments along the A-P axis. Since misexpression of another signaling molecule, *wg*, under the control of the

same *twist-GAL4* driver, has no obvious effects on germ cell behavior Hh (Figure 2; see also Boyle et al., 1997), it would appear that the aberrant migration pattern seen in *twist-GAL4/UAS-hh* embryos is due to ectopic *hh* expression.

Defects in germ cell migration are also observed in older stage 14–15 *twist-GAL4/UAS-hh* embryos. In the examples shown in Figures 3B–3D, many of the germ cells are clustered near the very posterior of the embryo. If the ectopically expressed Hh is responsible for directing germ cell migration to the posterior, then the *twist-GAL4* driver should be active in this region of the embryo. This is the case. As can be seen from the  $\beta$ -galactosidase antibody staining pattern in Figure 3A, the *twist-GAL4* transgene drives high levels of *UAS-LacZ* expression at the posterior of stage 14–15 embryos.

To provide additional evidence that germ cells migrate toward ectopic sources of Hh, we used a second mesodermal driver, *24B*. As can be seen in the early stage 11 embryo shown in Figure 3E, *24B* activates a *UAS-LacZ* transgene in a region of the mesoderm just posterior to the midgut invagination (shorter arrow). Germ cells (longer arrow) are found to congregate in this same region of stage 11 *24B/UAS-hh* embryos (Figure 3F).

#### **The Misplaced Germ Cells Do Not Associate with Ectopic Clift-Positive SGP Cells**

One explanation for the germ cell migration defects evident when Hh is expressed by different *UAS-GAL-4* drivers is that this signaling molecule induces the de novo



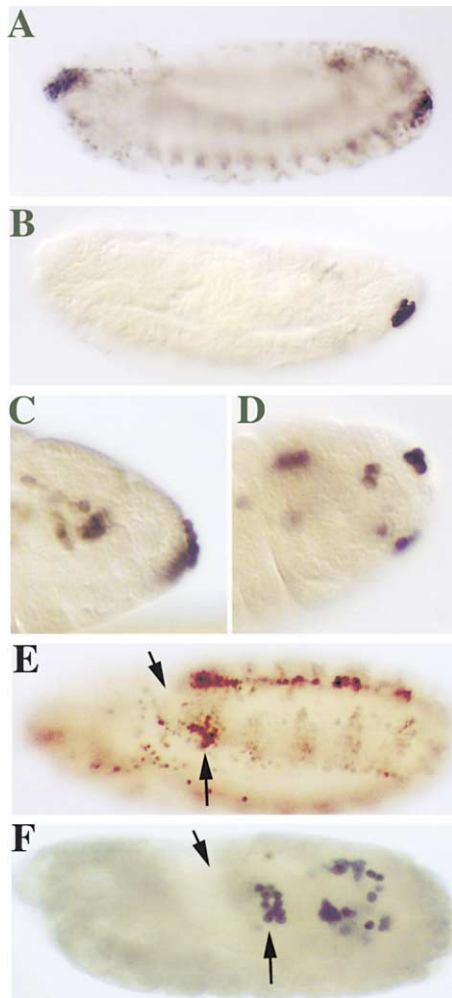


Figure 3. Germ Cells Migrate to Sites of Ectopic Hh

To determine where the *twist-Gal4* driver is active, *twist-GAL4/UAS-LacZ* embryos were stained with antibodies against  $\beta$ -galactosidase. (A) shows the  $\beta$ -galactosidase staining pattern of a stage 13–14 *twist-Gal4/UAS-LacZ* embryo. Note high levels of  $\beta$ -galactosidase in the posterior region of the embryo. (B)–(D) show *twist-GAL4/UAS-hh* stage 14–15 embryos stained with antibodies against Vasa. As can be seen in the photos, the germ cells in the *twist-GAL4/UAS-hh* embryos are often localized in one or a few clusters close to the posterior end, near the ectopic source of Hh protein.

To determine where the *24B-Gal4* driver is active, *24B-GAL4/UAS-LacZ* embryos were stained with antibodies against  $\beta$ -galactosidase. (E) shows the  $\beta$ -galactosidase staining pattern of a stage 11 *24B-Gal4/UAS-LacZ* embryo. Note high levels of  $\beta$ -galactosidase in the mesodermal precursor cells of the embryo, especially in the cells just to the right of the posterior midgut invagination (long arrow). (F) shows a slightly older *24B-GAL4/UAS-hh* embryo stained with antibodies against Vasa. A congregation of germ cells can be seen in the region of the mesoderm next to the midgut invagination (long arrow). Another cluster of germ cells is also seen just posterior to the first cluster. The third group of germ cells, which are located above and to the right of the second group have migrated to the expected position. In both the (E) and (F), the position of posterior midgut invagination is marked with shorter arrows.

formation of SGP cells at ectopic sites in the transgenic embryos. These ectopic SGP cells would then send attractive signals to the migrating germ cells, diverting them from the normal SGP cell clusters in PS10–12. To

test this possibility, we simultaneously stained embryos with Vasa to mark the germ cells and Clift antibody to mark the SGP cells (Boyle et al., 1997). In wild-type embryos, Vasa-positive germ cells are associated with the two bilateral clusters of Clift-positive SGP cells in stage 12–13 embryos. This is illustrated for one of these clusters in the first panel of Figure 4. Besides the SGP cells, there are other Clift-positive cells in wild-type embryos at this stage of development which are found in groups of 4–10 cells on either side of the ventral midline in each segment (not shown; see *UAS-hh/twist-GAL4* embryo). However, the germ cells do not normally interact with these other Clift-positive cells.

The second and third panels of Figure 4 show the Vasa and Clift staining pattern in two *UAS-hh/twist-GAL4* embryos. A subset of the germ cells in these embryos are found to be associated, as in wild-type, with the Clift-positive SGP cells in PS10–12. Germ cells at anomalous locations both anterior and posterior to the SGP cell clusters in PS10–12 are also evident; however, these misplaced germ cells are not associated with Clift-positive cells. This finding indicates that the ectopically expressed Hh protein does not induce the de novo formation of Clift-positive SGP cells. However, it can't be ruled out that ectopic expression of *hh* alters germ cell migration by creating a novel type of Clift negative SGP cells.

#### Germline Clones for Components of the Hedgehog Pathway

Two different models could potentially account for the effects of ectopic Hh on germ cell migration. First, ectopic Hh could be inducing a secondary signal, which in turn is responsible for altering germ cell migration. Alternatively, it is possible that germ cells respond directly to the Hh signaling molecule. In the latter case, cell-autonomous components of the *hh* signaling pathway should be required in the germline for proper migration. To test the involvement of the *hh* pathway, we generated germline clones (Chou and Perrimon, 1992) for mutations in two genes, *smoothened* (*smo*) (Alcedo et al., 1996; Van den Heuvel and Ingham, 1996) and *patched* (*ptc*) (Hooper and Scott, 1989; Nakano et al., 1989), which encode membrane proteins that function in the reception of the *hh* signal. *smo* is required to activate the signaling cascade in cells responding to Hh, and loss of *smo* activity has phenotypic effects similar to that of *hh* mutants (Van den Heuvel and Ingham, 1996). In contrast, *ptc* functions to antagonize the Hh signal, blocking the activation of *smo* (Chen and Struhl, 1996, 1998; Deneff et al., 2000). In *ptc* mutants, the signaling pathway is activated independent of the *hh* ligand.

Females carrying *smo* or *ptc* germline clones were mated to wild-type males, and we then examined segmentation and *wg* expression in the resulting embryos. As reported by Van den Heuvel and Ingham (1996), we found that the somatic development of embryos lacking maternal *smo* (*smo<sup>mat-</sup>*) can be fully rescued by a wild-type paternal copy of the gene (*smo<sup>zyg+</sup>*) (see Experimental Procedures). The pattern of *wg* expression in the *smo/+* offspring is indistinguishable from wild-type (not shown) and the heterozygous offspring develop into fertile adults with fully wild-type morphology. To confirm that the formation of the somatic gonad in the *smo<sup>mat-</sup>smo<sup>zyg+</sup>* embryos is normal, we probed with Clift anti-

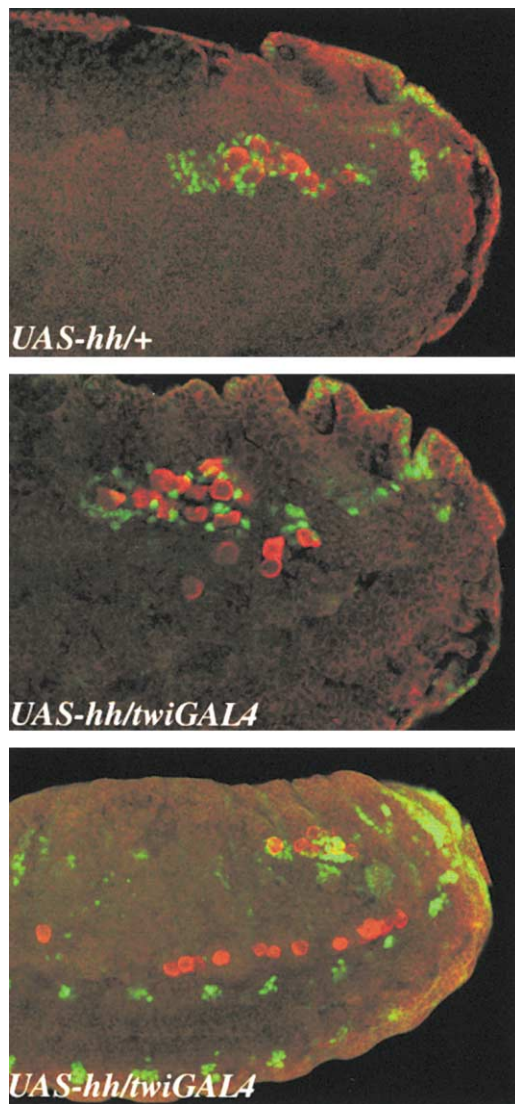


Figure 4. Ectopic Hh Does Not Induce the Formation of Clift-Positive SGPs

Hh was ectopically expressed in the lateral mesoderm using *twist-GAL4* driver. Homozygous *twist-GAL4* females were mated with homozygous *UAS-hh* males. Embryos derived from this cross were stained with anti-Clift (green) and anti-Vasa (red) antibody. In addition to somatic gonadal precursor cells, anti-Clift antibody labels other mesodermal cells in each segment. As a control, embryos derived from the *UAS-hh* stock alone were stained (top panel). As can be seen in the control *UAS-hh/+* embryo, virtually all of the germ cells are associated with the Clift-positive SGP cells in stage 13 embryos. In the experimental cross (*UAS-hh*  $\times$  *twi-GAL4*), germ cells are also found associated with Clift-positive SGP cells in the appropriate parasegments. However, there are also a large number of germ cells scattered in the posterior half of each embryo. The germ cells at ectopic sites are not associated with Clift-positive cells.

body. As can be seen in Figure 5C, the cluster of Clift-positive SGP cells in the *smo<sup>mat-</sup>-smo<sup>zyg+</sup>* embryos resembles that seen in wild-type embryos (Figure 5A). As was found for *smo*, embryos lacking maternal *ptc* (*ptc<sup>mat-</sup>*) can be fully rescued by a wild-type paternal copy of the gene (see Experimental Procedures), and in

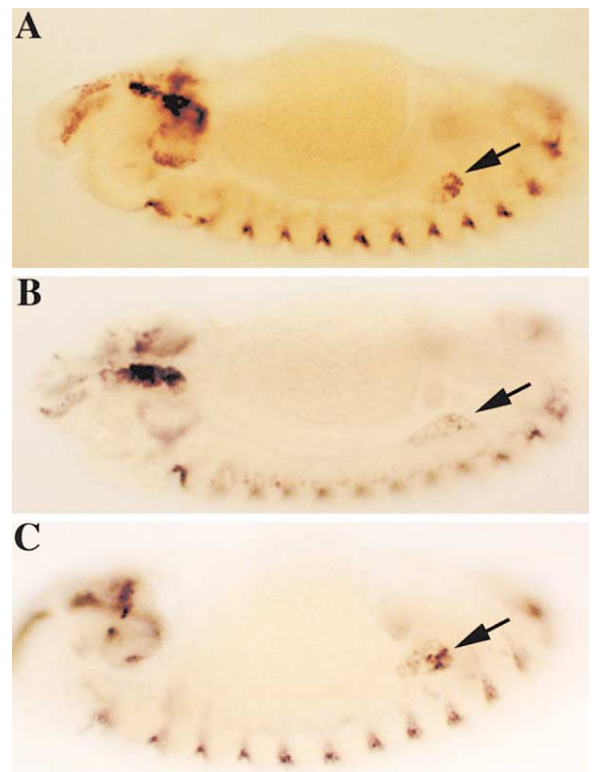


Figure 5. Formation of Somatic Gonad Is Normal in *Ptc<sup>mat-zyg+</sup>* and *smo<sup>mat-zyg+</sup>* Embryos

Mothers carrying germline clones for *smo* or *ptc* were mated with wild-type males and the progeny stained with anti-Clift antibody. (A) Wild-type embryo; (B) *Ptc<sup>mat-zyg+</sup>* embryo; (C) *smo<sup>mat-zyg+</sup>* embryo. Somatic gonadal precursor cells are formed normally in *Ptc<sup>mat-zyg+</sup>* and *smo<sup>mat-zyg+</sup>* embryos. No significant alterations with respect to morphology, position, or number were evident.

this case as well as the formation of the somatic gonad, it appears to be normal (Figure 5B).

The fact that *ptc<sup>mat-</sup>* and *smo<sup>mat-</sup>* embryos develop properly when fertilized by a wild-type sperm indicates that zygotic transcription of the paternal gene is sufficient to overcome any *hh* signaling defects in the soma resulting from the absence of the maternal gene product. However, unlike the soma, transcription in the germline is delayed until midway through embryogenesis. Consequently, we reasoned that the wild-type paternal gene might not be active for a sufficient time to fully compensate for cell-autonomous *hh* signaling defects in the germ cells of embryos lacking maternally derived *ptc* or *smo*.

***ptc* Germline Clones.** We first examined germ cell migration in embryos derived from *ptc* germline clones. For these experiments, mothers carrying *ptc* germline clones were mated to either *ptc/+* or wild-type fathers. In the former case, the severe segmentation defects in embryos lacking both maternal and zygotic *ptc* would be expected to disrupt germ cell migration independent of whether *ptc* function is required in the germ cells. For this reason, we only analyzed embryos (50% of total) that had a near wild-type *wg* expression pattern and consequently are expected to have zygotic *ptc* function. In the latter case, all embryos lack maternal *ptc* (*ptc<sup>mat-</sup>*);

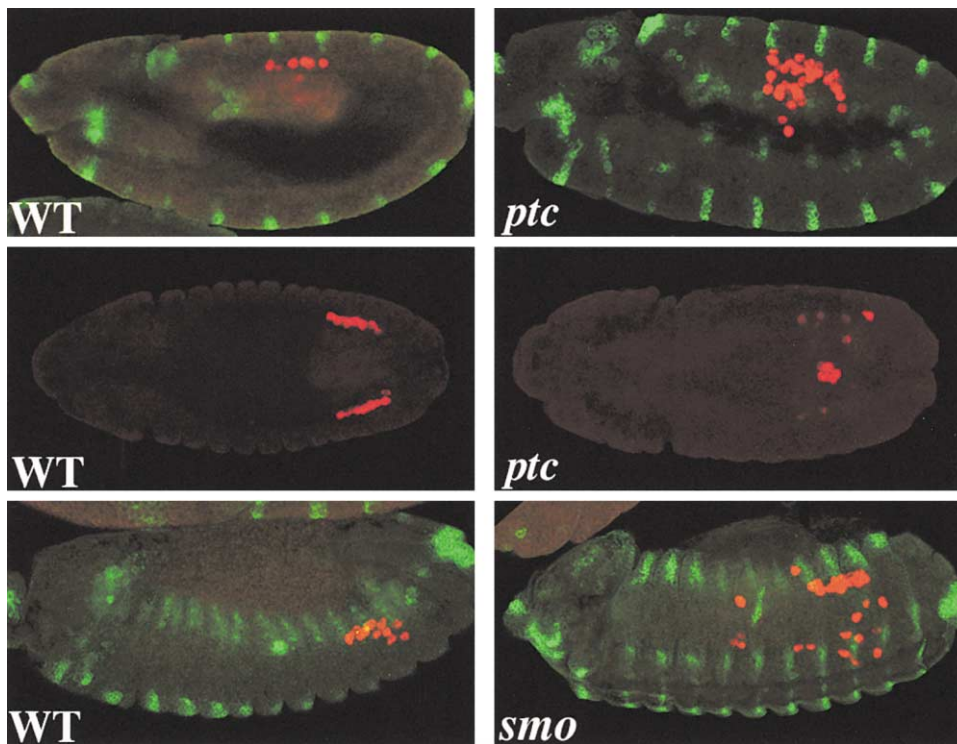


Figure 6. Germ Cells Lacking Maternal *ptc* or *smo*

Whole-mount antibody staining using Vasa (imaged in red) and Wingless (imaged in green) antibodies was performed on wild-type embryos and on *ptc*<sup>-/+</sup> or *smo*<sup>-/+</sup> embryos produced respectively by mating wild-type males to either *ptc* or *smo* germline clone mothers (labeled as *ptc* or *smo*). Up until stage 8–9, germ cell behavior in embryos lacking maternal *ptc* is indistinguishable from wild-type (not shown). However, as illustrated in the stage 10–11 *ptc* embryo shown in the top right panel, the number of germ cells is elevated. Wild-type: ~15 embryos counted: 21 germ cells/embryo. *Ptc*<sup>mat-zyg+</sup>: ~15 embryos counted: 28 germ cells/embryo. Additionally, the germ cells form small clumps, unlike embryos from wild-type mothers (>100 embryos of this stage examined and almost all the embryos showed premature germ cell clumping). In wild-type stage 13 embryos, ~8–10 germ cells contact the SGP cells in PS10–12 (middle left panel). In contrast, in stage 13 embryos lacking maternal *ptc*, fewer (2–3) germ cells are seen at the correct position, while several germ cell clumps can still be detected in the middle of the embryo (middle right panel). In embryos lacking maternal *smo*, germ cell migration resembles that in wild-type embryos until stages 12–13 (not shown). Around stage 13, germ cells in wild-type embryos begin coalescing with SGP cells to form the primitive embryonic gonad (bottom left). In embryos lacking maternal *smo*, many of the germ cells fail to sustain contact with the SGP cells and instead scatter through the posterior of the embryo (bottom right). Again, >100 embryos of this genotype were examined and all the embryos of this genotype displayed the phenotype. Note that the Wg antibody staining in the *smo* embryo is essentially the same as the wild-type embryo. This is as expected, since *smo*<sup>-/+</sup> embryos lacking maternal *smo* develop into normal adult animals. Later, around stage 15, the distribution of germ cells in *smo*<sup>mat-zyg+</sup> embryos resembles wild-type, indicating that the earlier defects are largely rescued. We suspect that this rescue is due to the zygotic activation of *smo* (see text). Unlike in the case of *Ptc*<sup>mat-zyg+</sup> embryos, no significant alteration in germ cell number was observed in *smo*<sup>mat-zyg+</sup> embryos. Wild-type: ~12 embryos counted: 20 germ cells/embryo. *smo*<sup>mat-zyg+</sup>: ~15 embryos counted: 21.4 germ cells/embryo.

however, because they have zygotic *ptc* activity, segmentation is normal. Essentially the same effects on germ cell development were obtained in the two crosses and these are summarized below.

The lack of maternal *ptc* has no apparent effect on germ cell formation and the number of pole cells is comparable to that in embryos from wild-type mothers (not shown). Pole cell migration into the interior of the embryo during gastrulation also appears to be normal. However, abnormalities become evident around stage 10–11 after the germ cells traverse through the midgut primordium and begin moving toward the dorsal surface of the endoderm. As shown in the panels on the right in Figure 6, many of the *ptc* germ cells prematurely associate, forming clumps of varying sizes, while others scatter to different sites through the mesoderm in the posterior half of the embryo. The clumps generally persist and can still be seen associated with the endoderm

in the middle of the embryo at stage 13, when the germ cells should have already migrated laterally through the mesoderm and interacted with the two bilateral clusters of SGPs cells (compare panels on left in Figure 6). Because of the premature clumping and the failure to migrate properly in the mesoderm, the primitive gonad in stage 15 embryos usually has fewer germ cells than are found in wild-type (not shown). Finally, germ cells from *ptc* mothers also seem to be defective in maintaining the cell cycle arrest at the G2/M transition. This is suggested by the unusually large number of germ cells evident in stage 9–10 embryos (see panels on right in Figure 6 and figure legend). It is interesting to note that the increase in germ cell number is only evident after they have traversed the midgut, while at earlier stages, the cell number is comparable to that in wild-type. This is in marked contrast to the effects of removing maternal *nanos* (*nos*), where the cell cycle defects are observed



as soon as the pole cells form at the blastoderm stage (Asaoka et al., 1999; Deshpande et al., 1999). This could indicate that *ptc* and *nos* use different mechanisms to attenuate the cell cycle.

***smo* Germline Clones.** As in the *ptc* experiments, we mated females with *smo* germline clones to either *smo*<sup>+/+</sup> or wild-type males. In the former case, embryos lacking both maternal and zygotic *smo* were identified on the basis of their abnormal *wg* expression pattern (50% of total) and excluded from further analysis. The remaining embryos, which lack only maternally derived *smo* activity, exhibited germ cell migration phenotypes similar to those observed when *smo*<sup>-</sup> mothers were mated to wild-type fathers.

As expected from their antagonistic relationship, the effects of removing maternal *smo* are quite distinct from those observed when maternal *ptc* is removed. Unlike the *ptc*<sup>mat-</sup> germ cells, *smo*<sup>mat-</sup> germ cells exhibited no apparent cell cycle defects, and the number of germ cells up until stage 14–15 is equivalent to that seen in wild-type (see legend to Figure 6). Similarly, the *smo*<sup>mat-</sup> germ cells traversed through the midgut and then migrated bilaterally along the mesoderm toward the SGP cells in parasegments 10–12 in a pattern much like that seen for wild-type germ cells. However, at stages 12–13, when the germ cells should begin establishing contacts with the SGP cells, clear-cut defects were observed in all embryos (Figure 6). Many of the *smo*<sup>mat-</sup> germ cells failed to associate with the SGPs and instead scattered randomly in the mesoderm, ending up at abnormal positions in more anterior or posterior parasegments. Later, at stage 15, these migration defects seem to be largely rectified and the number of *smo*<sup>mat-</sup> germ cells that coalesce into the primitive embryonic gonad approaches that seen in wild-type. We presume that *smo* transcription in the germ cells is responsible for rescuing the migration defects seen at stages 12–13.

#### Downstream Components of *hh* Pathway Are also Involved in Germ Cell Migration

The results described in the previous sections implicate two cell-autonomous components of the *hh* signaling pathway in germ cell migration. To provide further evidence that germ cells must receive and properly respond to an *hh* signal in order to migrate toward and make appropriate contacts with SGP cells, we tested two genes, *protein kinase A* (*pka*) (Lane and Kalderon, 1995) and *fused* (*fu*) (Preatet et al., 1990), which are downstream of *ptc* and *smo*, respectively. Like *ptc*, *pka* acts to antagonize the *hh* signal, and in *pka* mutants, the *hh* signaling pathway is inappropriately activated independent of the ligand. However, *ptc* and *pka* mutants are not precisely equivalent, and it is thought that there is a parallel pathway for controlling the transmission of the *hh* signal in the receiving cell that depends on *fu* rather than *pka*. *fu* functions within the receiving cell in the transmission of the signal, and *fu* mutants have phenotypic effects resembling *smo* (reviewed in Ingham, 1998; Johnson and Scott, 1998).

If germ cell migration depends on the reception and response to an *hh* signal, then removal of maternal *pka* activity should have phenotypic effects that are similar to those seen in *ptc* mutants, while removal of maternal *fu* activity should have phenotypic effects equivalent to those seen in *smo* mutants. This is the case. As shown

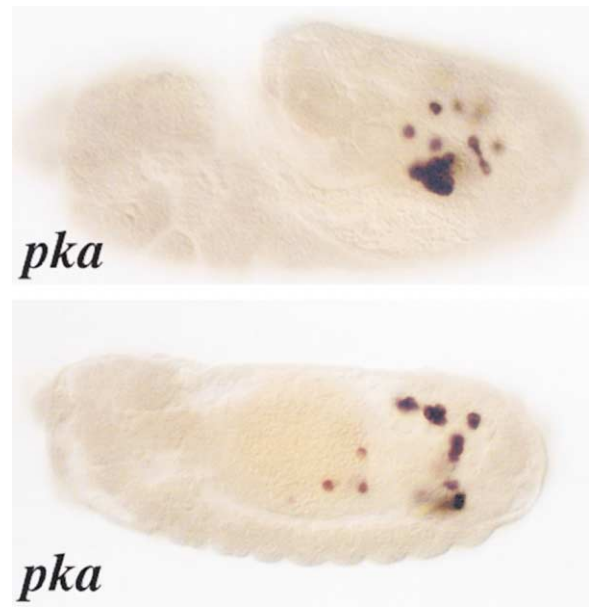


Figure 7. Germ Cells Lacking Maternal *pka* Clump Prematurely

Whole-mount antibody staining using anti-Vasa antibody was performed on *pka*<sup>+/+</sup> embryos produced by *pka* germline clone mothers. As in embryos lacking *ptc*, germ cells in the *pka* embryos clump into small clusters beginning around stage 10 (top panel, roughly 50 embryos examined). Small clumps of varying sizes persist in stage 13 embryos (bottom panel, >30 embryos). Note that unlike the results for either *ptc* or *smo* germline clones, the paternal wild-type copy of the *pka* gene does not rescue the somatic defects that arise in the absence of maternal *pka* and these *pka*<sup>+/+</sup> embryos do not hatch. This is also true for *fu*.

in Figure 7, germ cells lacking maternal *pka* exhibit migration and developmental defects similar to those seen in *ptc* mutants. The *pka*<sup>mat-</sup> germ cells prematurely associate into clumps in stage 11 embryos and then, at later stages, remain in these clumps in the center of the embryo instead of migrating bilaterally toward the SGPs. Moreover, like embryos produced from *ptc* mutant germline clones, there is an increase in germ cell number indicative of a failure to maintain cell cycle arrest. *fu* germline clones give germ cell migration defects equivalent to those seen in the absence of maternal *smo* function: the *fu*<sup>mat-</sup> germ cells migrate toward the SGPs, but fail to properly associate with these soma cells and instead scatter through the posterior half of the embryo (not shown). It should be noted that *pka* and *fu* germline clones differ from *ptc* and *smo* germline clones in that somatic development is not fully rescued by a wild-type paternal gene. Thus, even though the germ cell migration defects seen in *pka*<sup>mat-zyg+</sup> and *fu*<sup>mat-zyg+</sup> embryos closely resemble those found in *ptc*<sup>mat-zyg+</sup> and *smo*<sup>mat-zyg+</sup> embryos, respectively, it is possible that upsets in somatic patterning also contribute to the problems seen in germ cell migration.

#### Discussion

Previous studies have implicated both cell-autonomous and nonautonomous factors in germ cell migration. Two of the better characterized cell-autonomous factors are

the Nanos (Nos) protein and its regulatory partner, Pumilio (Pum) (Asaoka et al., 1999; Deshpande et al., 1999). Pole cells in progeny of mothers lacking either *nos* or *pum* activity do not migrate properly and fail to associate with SGP cells to form the primitive gonad. In the case of *nos*, abnormalities in germ cell migration are observed even when the somatic segmentation defects are corrected by the removal of maternal *hunchback* mRNA (Forbes and Lehmann, 1998). It is thought that the failure to establish/maintain transcriptional quiescence in germ cells, which normally occurs in the precellular blastoderm, is likely to be the major cause of the migration defects in *nos* and *pum* embryos. Factors that function nonautonomously in germ cell migration have been identified in genetic screens (Zhang et al., 1996; Moore et al., 1998b). Many of these correspond to well known patterning genes such as *eve*, *tin*, and *abd-A*, and the germ cell migration defects in mutant embryos are thought to arise largely because of defects in mesoderm development, a failure in the specification of SGP cells, or other gross developmental abnormalities. While these patterning genes are, for the most part, far removed from the molecules that actually signal germ cells and direct their movement, there are two loci, *wunen* (*wun*) and *Columbus* (*clb*), which seem to have a much more intimate connection to this process (Zhang et al., 1997; Van Doren et al., 1998).

*wun* encodes a Type 2 phosphatidic acid phosphatase. The Wun protein is predicted to span the membrane 5 or 6 times and is most probably involved in some aspect of lipid metabolism. In *wun* mutants, germ cells initially remain within the gut instead of migrating through the gut to the mesoderm. At later stages, the germ cells move at random, associating with mesodermal cells throughout much of the posterior half of the embryo. When *wun* is ectopically expressed in the mesoderm, it appears to block germ cell migration, and most germ cells remain in the yolk. These findings, together with its dynamic expression pattern in the endoderm and mesoderm, have led to the suggestion that *wun* is required for the production/activity of a repulsive signal which directs germ cells away from somatic cells that express Wun protein (Zhang et al., 1997). The other gene implicated in germ cell migration, *clb*, has the opposite function, controlling the production/activity of an attractive signal. Like *wun*, *clb* encodes an enzyme involved in lipid metabolism, HMG-CoA reductase. It catalyzes the synthesis of mevalonate, a precursor of cholesterol and isoprenoids (Van Doren et al., 1998). However, it is not known whether this particular biosynthetic pathway provides a precursor for the production of an attractant, or if Clb has some other unrelated function.

Beside *wun* and *clb*, little else is known about the somatic signals that attract/repel germ cells, or how these signals guide germ cell migration. We show here that the *hh* signaling pathway plays a critical role in the migration process. While previous studies have implicated *hh* in the formation of the somatic gonad and shown that germ cell migration is severely disrupted in *hh* mutant embryos, it was thought that the migration defects were a consequence of the misspecification of SGP cells (Moore et al., 1998b). Though this supposition is clearly correct, our results suggest that the Hh protein

also functions as a signaling molecule. Two lines of evidence support this view.

The first is the finding that ectopically expressed Hh protein induces germ cells to migrate to anomalous locations in the embryo. For all four of the GAL4 drivers tested, we found that the germ cells migrated toward the source of ectopic Hh; however, the most effective were the two drivers, *twist* and *24B*, which activate Hh expression at inappropriate sites in the mesoderm. This ability to alter the normal migration pattern and attract germ cells to sites of Hh synthesis is consistent with the idea that Hh functions as a nonautonomous signaling molecule in the germ cell migration pathway. While the simplest hypothesis is that Hh itself is the attractant, we can not exclude a more complicated model in which Hh is able to induce a variety of ectodermal and mesodermal cell types to synthesize some other unknown molecule that actually functions as the migration ligand.

The second is the finding that known cell-autonomous components of the Hh signaling pathway appear to be required in germ cells for normal migration behavior. Germline clones were used to test four different *hh* pathway genes—*ptc*, *pka*, *smo*, and *fu*. For all four, abnormalities in germ cell migration were observed in the progeny. In the case of both the *ptc* and *smo* germline clones, eggs fertilized by wild-type sperm developed into completely normal adults. Moreover, there are no apparent defects in the formation of the somatic gonad or in the pattern of Clift expression. These findings would support the view that the migration defects seen in *ptc<sup>mat-zyg+</sup>* and *smo<sup>mat-zyg+</sup>* embryos arise from cell-autonomous deficiencies in the response to Hh by the germ cells. However, it should be pointed out that there could be some undetected nonautonomous problem in somatic *hh* signaling in these embryos that induces abnormalities in germ cell behavior.

As would be expected from the known properties of these four genes in other well characterized *hh* pathways, the phenotypes produced by *ptc* and *pka* germline clones are similar and quite distinct from those observed for *smo* and *fu*. Moreover, the migration defects observed in *ptc/pka* and *smo/fu* germline clones can be explained by the antagonistic role of these genes in the *hh* signaling pathway. In the absence of maternal *ptc* or *pka*, *smo* and its downstream effectors in the *hh* pathway are activated in the germ cells independent of the Hh ligand. As a consequence, many of the germ cells clump together as they begin passing through the midgut, and then remain in place instead of migrating toward the SGP cells. Additionally, the mitotic cycle in *ptc<sup>mat-</sup>* (and to a lesser extent *pka<sup>mat-</sup>*) germ cells is inappropriately activated. Up regulation of cell division has been observed in somatic tumors that lack *ptc* function (Taipale et al., 2000) and in *ptc* mutant *C. elegans* germ cells (Kuwabara et al., 2000). In the case of *smo* and *fu*, the germ cells can't respond to the Hh ligand, and they are unable to detect or associate with the SGP cells, and instead migrate randomly through the mesoderm.

While these results indicate that maternally derived components of the *hh* pathway are important for germ cell migration, it should be noted that, since eggs from germline *ptc* and *smo* clones fertilized by wild-type sperm develop into *fertile* adults, at least some of the



germ cells must associate with SGP cells and form a functional gonad. This could mean that the *hh* pathway is important but not essential because there is some other, partially redundant, signaling pathway that can direct germ cell migration. While we can't rule out this possibility, we favor the idea that the maternal *hh* signaling defect in some of the germ cells can be rescued by the paternal wild-type copy of the signaling gene as it is in all of the somatic cells of these embryos. Consistent with this idea, transcription in germ cells is thought to commence around the time they start to exit the gut (Kobayashi et al., 1996). Since defects in migration first become evident shortly thereafter, we would suggest that rescue is incomplete because there is not quite enough time to synthesize sufficient quantities of the missing maternal gene product. By contrast, transcription begins much earlier in the soma and the lack of maternal gene product has no apparent phenotypic consequences.

In patterning, the Hh ligand can function as both a short and long distance signal (Chen and Struhl, 1996; Strigini and Cohen, 1997; Struhl et al., 1997). This raises the question of what role it plays in germ cell migration. It is possible that Hh acts only over a short distance, attracting germ cells when they are in close proximity to the SGP cells and/or mediating the direct interaction between the two cell types. Alternatively, the Hh ligand could act over a long distance and begin orchestrating germ cell migration as soon as the cells pass through the midgut. While not conclusive, several observations seem to favor this second alternative. First, migration defects in the absence of both maternal *ptc* and *pka* are evident at the time germ cells begin exiting the midgut. This argues that the machinery required to process and respond to the *hh* signal is most probably already present before the germ cells come in firm contact with the SGP cells. Second, *hh* expression driven by the *hairy-Gal4* driver induces aberrant germ cell migration at this same point in development, directing germ cells exiting the midgut to move ventrally rather than dorsally. The idea that Hh can act over long distances is also supported by the effects of ectopic Hh on the distribution of germ cells at later stages, when they would normally be associated with the SGP cells. Both the *elav-GAL4* and *twist-GAL4* drivers are able to attract germ cells to sites quite distant from the SGP cells in PS9–12. Moreover, the ability of ectopic Hh sources to attract germ cells seems to depend upon a competition with Hh protein produced by the endogenous *hh* gene. This is suggested by the finding that the migration defects induced by GAL4-UAS-driven expression of Hh protein are exacerbated when the embryos are heterozygous for a *hh* mutation (G.D., unpublished data).

The idea that Hh functions as a long distance signal for germ cell migration poses a number of problems. One is the question of specificity. Is the combination of an attractive Hh signal and a Wun-dependent repulsive signal sufficient to guide the complex movements of the germ cells after they exit the posterior midgut—they first move to the dorsal surface of the endoderm and then split into two groups which migrate toward the lateral mesoderm on either side of the embryo, where they find and associate with the SGP cells? The specificity problem is compounded by the fact that there are many

potential sources of Hh protein in the embryo. In the ectoderm, Hh is expressed in a stripe pattern in each parasegment (Heemskerk and Dinardo, 1994), while in the mesoderm, our data suggests that it is expressed not only in SGP cells, but also in the fat body precursor cells. If Hh protein emanating from these different sources is able to reach the germ cells at any point in their migration toward the SGP, the germ cells could be diverted toward inappropriate targets. In this case, one would have to suppose that some other signal, superimposed upon the Hh signal, is required to attract germ cells to the SGPs, and prevent them from being directed toward extraneous sources of Hh.

It is also possible that there are mechanisms which restrict or promote the movement of the Hh ligand within the embryo (Ramirez-Weber et al., 2000). For example, directional secretion of Hh protein coupled with a diffusion or transport barrier (such as the extracellular matrix) that seals the ectoderm off from the mesoderm could ensure that the Hh protein expressed in ectodermal cells only effectively signals other cells in the ectoderm. Within the mesoderm, there could be differences in the activity/mobility of Hh protein expressed in fat body and somatic gonadal precursor cells. In this context, it is interesting to note that *clb* expression in the mesoderm becomes progressively restricted to the SGP cells as the germ cells begin their migration. Although it is not known how *clb* functions in germ cell migration, an intriguing possibility is that it plays some role in the production or relay of the *hh* signal. The Hh protein is palmitoylated at the N terminus and has a cholesterol modification at the C terminus (Porter et al., 1996a, 1996b). Moreover, both of these modifications are believed to be important in its function as a long distance signaling molecule. Because *clb* encodes an enzyme which produces an intermediate for lipid biosynthesis, it would be reasonable to suppose that it might have a role in Hh modification. If this were the case, Hh protein expressed in SGP cells would be appropriately modified to function in long distance signaling, while Hh expressed in the fat body precursor cells would not.

Another important question is how the Hh signal actually promotes germ cell movement. In somatic cells, Hh signaling activates the Cubitus interruptus (Ci) transcription factor (Aza-Blanc et al., 1997; Wang and Holmgren, 2000; Methot and Basler, 2001). At this point, it is not clear whether Ci is the target of the Hh ligand in the germ cells. Moreover, if it is, how would the transcription of Ci target genes direct either germ cell migration or the association with SGP cells? Further studies will be required to resolve this and other issues.

## Experimental Procedures

### Strains and Culturing

Flies were grown on a standard medium at 25°C unless otherwise noted.

### Immunohistochemistry

The embryo stainings were performed essentially as described in Deshpande et al., 1995. Anti-Wingless and anti-Engrailed are both mouse monoclonal antibodies and were used at 1:10 dilution. Anti-Ci (rat) and anti-Vasa (rabbit) are polyclonal antibodies. Both the antibodies were preabsorbed against WT embryonic samples and were subsequently used at 1:500 dilution. Anti-β-galactosidase (rab-

bit) antibody was purchased from Kappel and was used at 1:1000 dilution after preabsorbing it against wild-type embryonic sample.

#### Misexpression Analysis

The following *UAS-Gal4* stocks were used for the misexpression studies: *UAS-hh*, *UAS-Wg*, *hairy Gal4*, *nanos Gal4*, *elav-Gal4*, and *twist Gal4*.

Typically, homozygous males carrying the *UAS-hh* or *UAS-wg* transgenes were mated with virgin females from the *Gal4* stocks. The resulting progeny embryos were fixed and stained for subsequent analysis. Homozygous *UAS-hh* males were mated with females carrying the relevant *GAL4* driver (*hairy* [*h*], *elav*, or *twist* [*twi*]) over a marked balancer chromosome. To visualize the migrating germ cells, we stained the embryos with anti-Vasa antibody. We compared the number of embryos showing defects in germ cell migration with those showing no migration defects. In each cross, approximately half of the total embryos are expected to show germ cell migration defects. Several hundred embryos from each cross were counted, and in all cases the expected 50/50 ratio was observed. To confirm our identification of embryos carrying both the *UAS-hh* and *GAL4* driver chromosomes, we repeated the same experiment, this time taking advantage of the marked balancer chromosomes which have either a *ftz-Lac-Z* or *Ubx-Lac-Z* transgene. We stained with both anti-Vasa and anti- $\beta$ -gal antibodies. As expected, the embryos that did not express  $\beta$ -galactosidase were the only embryos that showed obvious germ cell migration defects (data not shown). We also examined progeny from mothers homozygous for the *GAL4* driver and fathers homozygous for the *UAS-Hh* transgene. For all three drivers, greater than 70%–80% of the embryos showed germ cell migration defects (data not shown).

#### Germline Clonal Analysis

Germline clonal analysis was performed as described in Chou and Perrimon (1992), using the following stocks: *FRT-Patched* (*ptc<sup>ptcl</sup>*, obtained from Haifan Lin), *FRT-Smoothed* (*Smo<sup>smo2</sup>*, a cold sensitive amorph, obtained from G. Struhl); *FRT-fused* (*fu<sup>mtH63</sup>*); and *FRT-PKA* (*pka<sup>DCOH2</sup>*) (both obtained from D. Kalderon). Recombination was induced in the first or second instar larvae with a one hour heat shock at 37°C to activate *hs-flp*. Females carrying germline clones were mated with either wild-type males or males heterozygous for a mutation in the same gene. Embryos derived from these crosses were fixed and stained with the different antibodies for further analysis.

To confirm that we successfully generated *smo* and *ptc* germline clones, females carrying the clones were mated to *smo/+* or *ptc/+* males, respectively, and we then examined segmentation and *wg* expression in the offspring. For *smo*, two classes of offspring, each representing 50% of the total, were obtained. In the first class are embryos fertilized by wild-type sperm. Though they lack maternal *smo*, these embryos develop normally and become fertile adults. The second class are embryos fertilized by *smo* sperm. They lack both maternal and zygotic *smo* and exhibit severe segmentation defects. Like *hh* mutants, the cuticle of these *smo<sup>mat-zyg-</sup>* embryos is a lawn of denticles (Van den Heuvel and Ingham, 1996). The difference between the two classes can also be seen in the pattern of *wg* expression. Embryos lacking only maternal *smo* (*smo<sup>mat-</sup>*) have a near wild-type pattern of 14 *wg* stripes (*smo/+*). In contrast, *wg* expression is severely diminished in *smo<sup>mat-zyg-</sup>* embryos, just as it is in the absence of zygotic *hh* activity (*smo/smo*). Two equal classes of embryos were also observed for the *ptc* germline clones. One class, which we presume is fertilized by wild-type sperm, exhibited a nearly normal pattern of *wg* expression (*ptc/+*) and developed into fertile adults. The other class, which presumably lacks both maternal and zygotic *ptc*, showed severe segmentation defects. Consistent with the antagonistic relationship between *ptc* and *smo*, these *ptc<sup>mat-zyg-</sup>* embryos have a naked cuticle instead of the lawn of denticles seen for *smo<sup>mat-zyg-</sup>* embryos. Similarly, rather than being diminished, *wg* expression is substantially enhanced and much broader stripes are observed.

#### Acknowledgments

We thank Ira Clark, Steve DiNardo, Dan Kalderon, Paul Lasko, Haifan Lin, Roel Nusse, Norbert Perrimon, Trudi Schupbach, Gary Struhl,

and the Bloomington Stock Center for various fly stocks and antibodies. We are particularly grateful to Trudi Schupbach, Eric Wieschaus, and Shirley Tilghman for advice throughout the course of this work and comments on the manuscript. We would like to thank Joe Goodhouse for help with the confocal microscopy. Gordon Gray provided fly food. This work was supported by an NIH grant to P.S.

Received February 28, 2001; revised August 10, 2001.

#### References

- Alcedo, J., and Noll, M. (1997). Hedgehog and its patched-smoothed receptor complex: a novel signaling mechanism at the cell surface. *J. Biol. Chem.* **378**, 583–590.
- Alcedo, J., Ayzenzon, M., Von Ohlen, T., Noll, M., and Hooper, J.E. (1996). The *Drosophila* smoothed gene encodes a seven-pass membrane protein, a putative receptor for the hedgehog signal. *Cell* **86**, 221–232.
- Asaoka, M., Yamada, M., Nakamura, A., Hanyu, K., and Kobayashi, S. (1999). Maternal *pumilio* acts together with *Nanos* in germline development in *Drosophila* embryos. *Nature Cell Biol.* **1**, 431–437.
- Aza-Blanc, P., Ramirez-Weber, F.A., Laget, M.P., Schwartz, C., and Komberg, T.B. (1997). Proteolysis that is inhibited by hedgehog targets Cubitus interruptus protein to the nucleus and converts it to a repressor. *Cell* **89**, 1043–1053.
- Azpiaz, N., and Frasch, M. (1993). tinman and bagpipe: two homeo box genes that determine cell fates in the dorsal mesoderm of *Drosophila*. *Genes Dev.* **7**, 1325–1340.
- Azpiaz, N., Lawrence, P.A., Vincent, J.P., and Frasch, M. (1996). Segmentation and specification of the *Drosophila* mesoderm. *Genes Dev.* **10**, 3183–3194.
- Boyle, M., and DiNardo, S. (1995). Specification, migration and assembly of the somatic cells of the *Drosophila* gonad. *Development* **121**, 1815–1825.
- Boyle, M., Bonini, N., and DiNardo, S. (1997). Expression and function of clift in the development of somatic gonadal precursors within the *Drosophila* mesoderm. *Development* **124**, 971–982.
- Brand, A.H., and Perrimon, N. (1993). Targeted gene expression as a means of altering cell fates and generating dominant phenotypes. *Development* **118**, 401–415.
- Brohier, H.T., Moore, L.A., Van Doren, M., Newman, S., and Lehmann, R. (1998). *zfh-1* is required for germ cell migration and gonadal mesoderm development in *Drosophila*. *Development* **125**, 655–666.
- Brookman, J.J., Toosy, A.T., Shashidhara, L.S., and White, R.A. (1992). The 412 retrotransposon and the development of gonadal mesoderm in *Drosophila*. *Development* **116**, 1185–1192.
- Chen, Y., and Struhl, G. (1996). Dual roles for patched in sequestering and transducing Hedgehog. *Cell* **87**, 553–563.
- Chen, Y., and Struhl, G. (1998). In vivo evidence that Patched and Smoothed constitute distinct binding and transducing components of a Hedgehog receptor complex. *Development* **125**, 4943–4948.
- Chen, C.H., von Kessler, D.P., Park, W., Wang, B., Ma, Y., and Beachy, P.A. (1999). Nuclear trafficking of Cubitus interruptus in the transcriptional regulation of Hedgehog target gene expression. *Cell* **98**, 305–316.
- Chou, T.B., and Perrimon, N. (1992). Use of a yeast site-specific recombinase to produce female germline chimeras in *Drosophila*. *Genetics* **131**, 643–653.
- Chuang, P.T., and Komberg, T.B. (2000). On the range of hedgehog signaling. *Curr. Opin. Genet. Dev.* **10**, 515–522.
- Denef, N., Neubuser, D., Perez, L., and Cohen, S.M. (2000). Hedgehog induces opposite changes in turnover and subcellular localization of patched and smoothed. *Cell* **18**, 521–531.
- Deshpande, G., Stuke, J., and Schedl, P. (1995). *scute* (*sis-b*) function in *Drosophila* sex determination. *Mol. Cell Biol.* **15**, 4430–4440.
- Deshpande, G., Calhoun, G., Yanowitz, J.L., and Schedl, P.D. (1999). Novel functions of nanos in downregulating mitosis and transcrip-

- tion during the development of the *Drosophila* germline. *Cell* 99, 271–281.
- Ephrussi, A., and Lehmann, R. (1992). Induction of germ cell formation by *Oskar*. *Nature* 358, 387–392.
- Forbes, A., and Lehmann, R. (1998). *Nanos* and *Pumilio* have critical roles in the development and function of *Drosophila* germline stem cells. *Development* 125, 679–690.
- Frasch, M. (1995). Induction of visceral and cardiac mesoderm by ectodermal Dpp in the early *Drosophila* embryo. *Nature* 374, 464–467.
- Heemskerk, J., and DiNardo, S. (1994). *Drosophila* hedgehog acts as a morphogen in cellular patterning. *Cell* 76, 449–460.
- Hooper, J.E., and Scott, M.P. (1989). The *Drosophila* patched gene encodes a putative membrane protein required for segmental patterning. *Cell* 59, 751–765.
- Howard, K. (1998). Attractive genetics. *Nature* 396, 406–407.
- Jaglarz, M.K., and Howard, K.R. (1995). The active migration of *Drosophila* primordial germ cells. *Development* 121, 3495–3503.
- Ingham, P.W. (1998). Transducing Hedgehog: the story so far. *EMBO J.* 17, 3505–3511.
- Johnson, R.L., and Scott, M.P. (1998). New players and puzzles in the Hedgehog signaling pathway. *Curr. Opin. Genet. Dev.* 18, 450–456.
- Kalderon, D. (2000). Transducing the Hedgehog signal. *Cell* 103, 371–374.
- Kobayashi, S., Yamada, M., Asaoka, M., and Kitamura, T. (1996). Essential role of the posterior morphogen *nanos* for germline development in *Drosophila*. *Nature* 380, 708–711.
- Kuwabara, P.E., Lee, M.H., Schedl, T., and Jefferis, G.S. (2000). A *C. elegans* patched gene, *ptc-1*, functions in germ-line cytokinesis. *Genes Dev.* 15, 1933–1944.
- Lane, M.E., and Kalderon, D. (1995). Localization and functions of protein kinase A during *Drosophila* oogenesis. *Mech. Dev.* 49, 191–200.
- Luer, K., Urban, J., Klambt, C., and Technau, G.M. (1997). Induction of identified mesodermal cells by CNS midline progenitors in *Drosophila*. *Development* 124, 2681–2690.
- Methot, N., and Basler, K. (2001). An absolute requirement for Cubitus interruptus in Hedgehog signaling. *Development* 128, 733–742.
- Moore, L.A., Broihier, H.T., Van Doren, M., and Lehmann, R. (1998a). Gonadal mesoderm and fat body initially follow a common developmental path in *Drosophila*. *Development* 125, 837–844.
- Moore, L.A., Broihier, H.T., Van Doren, M., Lunsford, L.B., and Lehmann, R. (1998b). Identification of genes controlling germ cell migration and embryonic gonad formation in *Drosophila*. *Development* 125, 667–678.
- Nakano, Y., Guerrero, I., Hidalgo, A., Taylor, A., Whittle, J.R., and Ingham, P.W. (1989). A protein with several possible membrane-spanning domains encoded by the *Drosophila* segment polarity gene *patched*. *Nature* 341, 508–513.
- Porter, J.A., Ekker, S.C., Park, W.J., von Kessler, D.P., Young, K.E., Chen, C.H., Ma, Y., Woods, A.S., Cotter, R.J., Koonin, E.V., and Beachy, P.A. (1996a). Hedgehog patterning activity: role of a lipophilic modification mediated by the carboxy-terminal autoprocessing domain. *Cell* 86, 21–34.
- Porter, J.A., Young, K.E., and Beachy, P.A. (1996b). Cholesterol modification of hedgehog signaling proteins in animal development. *Science* 274, 255–259.
- Preat, T., Therond, P., Lamour-Isnard, C., Limbourg-Bouchon, B., Tricoire, H., Erk, I., Mariol, M.C., and Busson, D. (1990). A putative serine/threonine protein kinase encoded by the segment-polarity fused gene of *Drosophila*. *Nature* 347, 87–89.
- Ramirez-Weber, F.A., Casso, D.J., Aza-Blanc, P., Tabata, T., and Kornberg, T.B. (2000). Hedgehog signal transduction in the posterior compartment of the *Drosophila* wing imaginal disc. *Mol. Cell* 6, 479–485.
- Riechmann, V., Rehorn, K.P., Reuter, R., and Leptin, M. (1998). The genetic control of the distinction between fat body and gonadal mesoderm in *Drosophila*. *Development* 125, 713–723.
- Sam, S., Leise, W., and Hoshizaki, D.K. (1996). The serpent gene is necessary for progression through the early stages of fat-body development. *Mech. Dev.* 60, 197–205.
- Seydoux, G., and Strome, S. (1999). Launching the germline in *Caenorhabditis elegans*: regulation of gene expression in early germ cells. *Development* 126, 3275–3283.
- Smith, J.L., Wilson, J.E., and Macdonald, P.M. (1992). Overexpression of *oskar* directs ectopic activation of *nanos* and presumptive pole cell formation in *Drosophila* embryos. *Cell* 70, 849–859.
- Staehling-Hampton, K., Hoffmann, F.M., Baylies, M.K., Rushton, E., and Bate, M. (1994). dpp induces mesodermal gene expression in *Drosophila*. *Nature* 372, 783–786.
- St. Johnston, D. (1993). Pole plasm and the posterior group genes. In *The Development of Drosophila melanogaster*. (Cold Spring Harbor Laboratory Press: Cold Spring Harbor, NY). pp. 325–363.
- Strigini, M., and Cohen, S.M. (1997). A Hedgehog activity gradient contributes to AP axial patterning of the *Drosophila* wing. *Development* 124, 4697–4705.
- Struhl, G., Barbash, D.A., and Lawrence, P.A. (1997). Hedgehog acts by distinct gradient and signal relay mechanisms to organise cell type and cell polarity in the *Drosophila* abdomen. *Development* 124, 2155–2165.
- Taipale, J., Chen, J.K., Cooper, M.K., Wang, B., Mann, R.K., Milenkovic, L., Scott, M.P., and Beachy, P.A. (2000). Effects of oncogenic mutations in *Smoothed* and *Patched* can be reversed by cyclopamine. *Nature* 406, 1005–1009.
- Van den Heuvel, M., and Ingham, P.W. (1996). *smoothed* encodes a receptor-like serpentine protein required for hedgehog signalling. *Nature* 382, 547–551.
- Van Doren, M., Broihier, H.T., Moore, L.A., and Lehmann, R. (1998). HMG-CoA reductase guides migrating primordial germ cells. *Nature* 396, 466–469.
- Wang, Q.T., and Holmgren, R.A. (2000). Nuclear import of cubitus interruptus is regulated by hedgehog via a mechanism distinct from Ci stabilization and Ci activation. *Development* 127, 3131–3139.
- Warrior, R. (1994). Primordial germ cell migration and the assembly of the *Drosophila* embryonic gonad. *Dev. Biol.* 166, 180–194.
- Williamson, A., and Lehmann, R. (1996). Germ cell development in *Drosophila*. *Annu. Rev. Cell Dev. Biol.* 12, 365–391.
- Zalokar, M., and Erk, I. (1976). Division and migration of nuclei during early embryogenesis of *Drosophila melanogaster*. *J. Microsc. Biol. Cell* 25, 97–106.
- Zhang, N., Zhang, J., Cheng, Y., and Howard, K. (1996). Identification and genetic analysis of *wunen*, a gene guiding *Drosophila melanogaster* germ cell migration. *Genetics* 143, 1231–1241.
- Zhang, N., Zhang, J., Purcell, K.J., Cheng, Y., and Howard, K. (1997). The *Drosophila* protein *Wunen* repels migrating germ cells. *Nature* 385, 64–67.